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Q1) Explain the different types of failures in a database system and classify them with examples.

A system may face different types of failures that can interrupt transaction execution or affect the stored data.

These failures are generally classified into transaction failures, process (or system) failures, media failures and communication failures. Each type has specific causes and recovery methods.

i) Transaction failures

A transaction failure occurs when a transaction cannot complete its execution successfully.

ex:

A banking transaction trying to withdraw more money than available may abort due to insufficient balance.

ii) System / process failure.

These failures occur when the database system stops due to the internal issues, but the secondary storage (disk data) remains safe.

ex:

A server suddenly restarts due to power loss while multiple transactions were running. RAM contents are lost, but disk data remains intact.

iii) Media failure.

Also called device failure or storage failure. It happens when the storage device containing the database is physically damaged.

ex:

A database stored on a hard drive becomes unreadable because the disk develops serious bad sectors. Recovery requires backups.

iv) Communication failure.

occurs in distributed or network-based database systems when connecting between sites or users are interrupted.

ex:

In an online shopping application, a transaction may fail and rollback. If internet connection is lost between the user and the server.

v) Application Software failure.

Failure caused by bugs in the application that interacts with the database.

ex:

A programming error in application incorrectly updates records, causing transaction rollback.

(Q2) Describe log-based recovery in DBMS, Explain the roles of UNDO and REDO operation in recovering database.

Log-based recovery:

- Log-based recovery is a technique used in DBMS to restore the database to a consistent state after a system crash or failure.
- The DBMS maintains a log file (transaction log) that records every write operation performed by transactions.
- Each log entry contains the transaction ID, the data item updated, its old value (before image) and new value (after image). According to the Write-Ahead Logging (WAL) rule, log records must be written to stable storage before the actual database is updated.
- During recovery, the system examines the log to identify the transactions that committed before the crash and those that were incomplete.

Two operations are used to bring the database back to a consistent state: UNDO and REDO.

UNDO

is applied to all uncommitted transactions. It uses the log's before images to restore the old values, ensuring incomplete transactions do not leave partial updates in the database.

REDO

is applied to committed transactions. It uses the after images stored in the log to reapply the updates of transactions that committed but whose changes might not have been written to

disk at the moment of crash.

Q3) Explain how a database can be recovered when multiple transactions are executing concurrently. Include the challenges faced during Concurrent transaction recovery.

- In a multiple transaction environment, the DBMS uses logs (write-ahead logging) to track all operation of each transaction.
- During recovery after a crash, the system scans the log to identify which transactions were committed, and which were uncommitted, at the moment of failure.
- REDO operations are applied for all committed transactions to ensure that their updates are permanently reflected in the database.
- UNDO operations are applied for all incomplete (concommitted) transaction to roll back their partial effect.
- Concurrency control techniques (locking, timestamps) ensure that the recovery process maintains isolation and consistency, even though transactions overlapped during execution.
- The checkpoint mechanism helps by reducing the amount of log that needs to be scanned during recovery.

Challenges in Concurrent Transaction Recovery:

- interleaving of operations.
- Maintaining transaction isolation.
- Cascading effects.
- Handling locks during recovery.
- performance overhead.

Q4) Describe database backup strategies and recovery mechanism for catastrophic failure. Explain the importance of remote backup systems and how they contribute to database security.

- Database backup strategies for catastrophic failure involve creating full ~~backup~~ differential, and incremental backups to ensure that the entire database and its recent changes can be restored after total system loss.

- Organizations also use cold backups for guaranteed consistency and hot backups to support 24x7 availability.

- During recovery the system restores the latest full backup, applies differential or incremental backup and then uses archived logs files to redo committed transactions and reconstruct the database to consistent state.

- For high-availability systems, standby or replicated databases allow quick failover during a disaster.

Remote backup system:

Plays a crucial role by storing database copies at geographically separate locations, protecting data from local disasters such as fire, hardware damage, or cyber attacks. They enhance database security through geographical redundancy, encrypted data transfer, isolated storage and versioned snapshots ensuring that even if the primary site is compromised, data can be safely recovered. This guarantees business continuity and minimize data loss during catastrophic failures.

Q5) Explain different database security issues.

Database security issues arise when an unauthorized user, malicious action, or system vulnerabilities threaten the confidentiality, integrity or availability of data.

Major issues include:

Unauthorized Access:

gaining access without proper authentication leading to data misuse.

SQL Injection Attack.

Injecting malicious SQL code to read or modify sensitive data.

Privilege Abuse:

Authorized users misusing excessive privileges for harmful actions.

Data Exposure:

Sensitive data revealed due to unencrypted storage or insecure backups.

Malware Attacks:

Malicious software encrypts or corrupts database files.

Weak Authentication:

Simple password and poor authentication policies allow brute force attacks.

Insider Threats:

Employee or admin accidentally or intentionally compromising data.

Denial of Service:

overloading the database server to make it unavailable.

Mis Configured:

Default settings, open ports, or improper roles exposing vulnerabilities.